

KIQ 2.1 — Final Takeaway

What emerging digital trends are shaping ordering behavior in the HoReCa sector?

Methodological note. Direct HoReCa-specific data is out of scope. Espacenet patent literature on digital ordering and AI-commerce technologies is used as a proxy — patents lead academic publications by 1–3 years, making them the canonical early-warning signal for KIT 2.

1. Patent activity is accelerating

Filing volume in digital ordering technologies grew **2.6× from 2022 to 2025** (111 → 288 patents/year). The trend is unambiguous: the technologies that will reshape ordering are being built *now*.

2. The technology stack is layered, not monolithic

IPC analysis reveals a coherent five-layer stack — commerce logic (G06Q), computing infrastructure (G06F), AI/ML (G06N), voice (G10L), and IoT (G16Y). For METRO, this means the digital ordering channels of the next 3–5 years won't be re-platformed once; they will be progressively augmented across the stack.

3. IP ownership is external and geographically concentrated

Top filers (Baidu, Tencent, eBay, Rakuten, IBM, Mastercard) are Asian and US-headquartered tech and fintech players. European presence is essentially absent. METRO's absence is consistent with its role as a deployment-stage wholesaler, not a technology inventor — the strategic question is *which* upstream providers to partner with, license from, or acquire.

4. AI is the headline framing, but it splits into distinct paradigms

Bigram analysis shows that beneath the dominant "artificial intelligence" signal lie specific frontiers: **generative AI**, **intelligent agents**, **recommendation engines**, plus persistent secondary signals around **blockchain** (supply-chain provenance) and **cross-border commerce** (regulatory/logistics). Voice ordering and IoT-enabled autonomous replenishment appear as smaller but unmistakable 2024–2026 leading indicators.

5. The HoReCa-specific gap is itself a finding

No top filer is HoReCa-specific. The technologies are sector-neutral and not locked behind vertical-incumbent IP — a deployment window is still open for METRO to embed them into HoReCa ordering before consolidation hardens.

Bottom line

Patent intelligence reveals a layered digital ordering technology stack — commerce, computing, and AI dominate; voice, IoT, generative AI, and intelligent agents signal where the next wave of customer interaction is being built — with external vendors holding most of the IP and a clear deployment window still open for European wholesalers.

Bridge to KIQ 1.3

In KIQ 1.3 we found AI, ML, CRM, marketplaces sitting at the *technology periphery* of the retention discourse — enablers of satisfaction, trust, and experience. KIQ 2.1 zooms into that periphery and

shows it's accelerating, externally controlled, and stack-deep. The strategic question for METRO is no longer *whether* to adopt, but **which layers to build, partner for, or acquire**.

Methodological challenge — "Isn't AI dominant just because you searched for it?"

This is the right methodological challenge, and the honest answer has three parts.

1. Yes, partially — and it's a known feature of query-driven corpus analysis.

AI/ML terms were in the search query, so patents matching those terms are over-represented relative to a random patent sample. Acknowledge this openly. Pretending otherwise is the wrong move.

2. But the AI dominance isn't *only* a query artifact:

(a) AI/ML were the *broadest* query terms (deliberately, to capture the field). Specific terms like "*predictive ordering*", "*voice ordering*", "*autonomous replenishment*", "*recommendation engine*" were also in the query — yet AI overshadows them by 10×. If the query were equally weighting all terms, we'd expect a flatter distribution. The asymmetric dominance is *real signal*, not just query mechanics.

(b) The IPC analysis is partially independent — G06N (AI/ML) wasn't a search term; it's a classification assigned by patent examiners. Its high prominence is examiner-validated, not query-driven.

(c) The trend curve is query-independent — the 2.6× growth from 2022 to 2025 holds whether we slice by AI or by any other term. The acceleration is structural.

3. The honest framing for the report:

The query defined the *boundary* of the corpus, but inside that boundary the data still tells you about **relative weight, structure, growth, and concentration** — which are the actual KIQ 2.1 questions. The query doesn't predetermine that Baidu dominates, that generative AI is rising in 2024–2026, or that voice/IoT remain niche-but-present. Those findings are corpus-internal, not query-driven.

Live one-liner if the professor asks:

"The query defines the scope of what we're looking at, not the structure of what we find inside it — the dominance, growth rates, and concentration patterns are corpus-internal findings, not query artifacts."